

AMBROSE BLAY

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Education

Cornell University - Ithaca, NY

Graduating May 2027

Major: Computer Science

Relevant Coursework: Computer Systems Organization, Operating Systems, Intro to Analysis of Algorithms, Intro to Database Systems, Data Structures and Functional Programming, Object-oriented Programming and Data Structures, Discrete Maths, Linear Algebra, Intro to Machine Learning, Intro to Backend Development

Involvements: National Society of Black Engineers, Underrepresented Minorities in Computing, Cornell Astronomical Society, Cornell Association of CS Undergrads (Web Developer), Intramural Soccer

TA positions: Intro to Computing (FA2024, SP2025, SP2026), Calculus I (FA2024, SP2025)

Research: Bioacoustic Source Separation & Localization with two Acoustic Vector Sensors

Technical Skills

Languages: Rust, C, C++, Java, OCaml, Python, PHP, JavaScript, SQL, SPARQL

Frameworks: Flask, Express, Firebase, React, SQLAlchemy, Java Swing, Spring Boot, MongoDB, GitHub, Docker, Git

Experience

Software Engineering Intern, Meta | Seattle, WA

May 2026 – Aug. 2026

- Designed and built an end-to-end proactive content moderation pipeline for Messenger group chats that processes 200M+ events/day to detect policy-violating group names and images before they reach teen users.
- Built a new async processing path through Meta's ML integrity infrastructure, including serialization schemas, async job orchestration with privacy zone integration, and model configuration that routes content to text and image classifiers
- Implemented the full detection-to-enforcement loop: real-time event observation, ML classifier scoring, automated decision evaluation, and enforcement actions (removing minors from flagged chats, blocking joins to violating groups)

Software Engineer Intern, Amazon Web Services | Seattle, WA

May 2025 – Aug. 2025

- Implemented Leapfrog Triejoin, a worst-case optimal join algorithm, in NeptuneDB's C++ query engine to optimize complex graph query execution.
- Developed heuristics-based query planner to find worst-case optimal join plans, successfully identifying optimum plans for 12/14 LUBM benchmark queries.
- Built a test suite using LUBM and WatDiv benchmarks, achieving full coverage across diverse query graphs.
- Implemented a Java prototype comparing binary join plans with worst-case optimal join plans, demonstrating improved query performance for complex query graph patterns.

Software Engineer, Culture Care | Ithaca, NY

Dec. 2023 – May 2024

- Created react-based online forms, streamlining client onboarding and matching over 30 patients to BIPOC therapists
- Integrated Firebase for client-side authentication, protecting sensitive therapist and patient data.
- Implemented patient hard/soft pass algorithm, reducing practitioner appointment review time by approximately 60%.

Projects

Ambra Programming Language | C++17, CMake

<https://t.ly/hBP1e>

- Designed and implemented a compiled, statically typed programming language with custom lexer, recursive descent parser, type checker, and stack-based virtual machine executing intermediate representation instructions.
- Built multi-pass compiler pipeline with semantic analysis, IR lowering, and bytecode validation; includes 400+ unit tests using Google Test framework covering lexical analysis, parsing, semantic analysis, and IR lowering.

SyncFlix | Node.js, Express, React, MongoDB, Socket.io

<https://t.ly/rym0m>

- Built a real-time video streaming platform enabling multiple users to watch shows together with synchronized playback and group chat using WebSockets.
- Used MongoDB GridFS for efficient video bucket storage and JWT for secure user auth and session management.

Distributed Proxy System | Python, TCP Networking, Caching, Load Balancing

<https://t.ly/i2ehV>

- Built a distributed proxy layer with TTL/LRU caching and per-node metrics to improve latency and reduce origin load.
- Implemented round-robin and least-loaded load balancing with active health checks and failover for high availability.

UniNav | React, Flask, SQLAlchemy, BeautifulSoup

<https://t.ly/82MWz>

- Built a full-stack app showing real-time availability of 100+ lecture halls at Cornell.
- Scraped 10K+ class schedules into an SQL database and served data via Flask API. Deployed on Heroku

Languages & Interests

Languages: English, Twi (Native)

Additional Interests: Astrophotography, Soccer, Sports games, African Art